Does WTI Crude Oil Have Intraday Momentum?

A post-2020 replication of Baltussen, Da, Lammers and Martens (2021)

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FINM 37000, Group 3

What we test

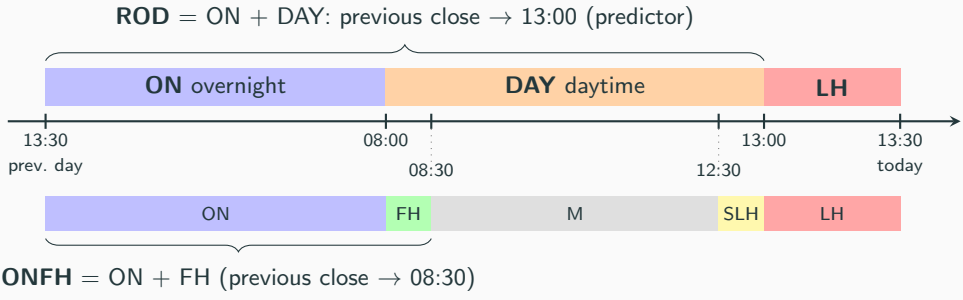
Claim (BDLM 2021): the market's move from yesterday's close until 30 minutes before today's close predicts the last 30 minutes.

- **LH**: return 13:00–13:30 Chicago time (what we predict)
- **ROD**: return from previous 13:30 to 13:00 (the predictor)
- **ON** / **DAY**: ROD split at 08:00 into overnight and daytime parts
- **Slope**: basis points of LH per 1% of the predictor. Positive = momentum, negative = reversal.

Data: Databento CME one-minute bars, CL futures. 1,436 trading days after the paper's filters.

- **In-sample** (slopes, errors, intervals): all 1,436 days, 4 Jan 2021 to 7 Oct 2026, fitted once.
- **Out-of-sample** (R^2 vs. the historical mean): expanding window. The first 500 days (to Dec 2022) are used only for fitting. Then 936 daily forecasts, 4 Jan 2023 to 7 Oct 2026, each fitted on earlier days only.

Timeline of the predictors (Chicago time, not to scale)



FH: first half hour after the 08:00 open. **M**: 08:30 to 12:30. **SLH**: 12:30 to 13:00, the second-to-last half hour. The decision is made at 13:00: every predictor uses prices only up to then. $(1 + ON)(1 + DAY) = 1 + ROD$. 13:30 Chicago time (14:30 ET) is the reference close, not the end of the electronic session.

Replication: the paper's regressions

Each day t, regress the last-half-hour return on a predictor, with an intercept:

$$LH_t = \alpha + \beta\, ROD_t + \varepsilon_t$$

The paper also uses two other versions:

$$LH_t = \alpha + \beta\, ONFH_t + \varepsilon_t$$
$$LH_t = \alpha + \beta_1\, ONFH_t + \beta_2\, M_t + \beta_3\, SLH_t + \varepsilon_t$$

ONFH: overnight plus first half hour (previous close to 08:30). M: 08:30 to 12:30. SLH: 12:30 to 13:00.
Null: $\beta = 0$ (no predictability). $\beta > 0$ is momentum.
We report $\hat{\beta}$ with Newey-West standard errors (7 lags), plus expanding-window out-of-sample R^2 against the historical mean.

Our three new hypotheses

H1. Overnight vs. daytime.

$$LH_t = \alpha + \beta_{ON}\, ON_t + \beta_{DAY}\, DAY_t + \varepsilon_t, \qquad H_0 :\ \beta_{ON} = \beta_{DAY}$$

H2. Volatility regimes. $HV_t = 1$ if realized variance up to 13:00 is above the 75th percentile of earlier days.

$$LH_t = \alpha + \beta_{ON} ON_t + \beta_{DAY} DAY_t + \gamma_{ON}\, ON_t HV_t + \gamma_{DAY}\, DAY_t HV_t + \varepsilon_t$$

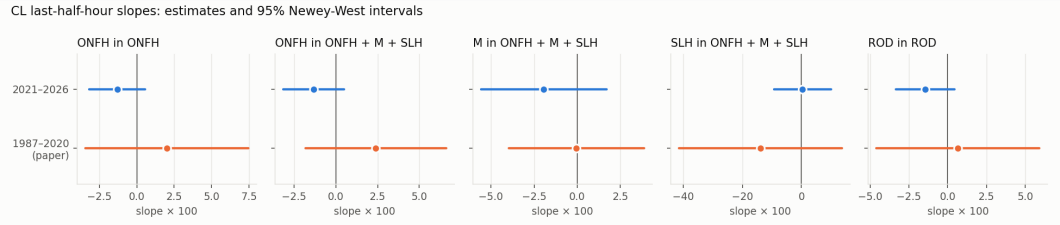
$$H_0 :\ \gamma_{ON} = \gamma_{DAY} = 0$$

H3. Agreement. $Agree_t = 1$ if ON_t and DAY_t have the same sign.

$$LH_t = \alpha + \beta\, ROD_t + \gamma\, Agree_t + \delta\, ROD_t Agree_t + \varepsilon_t, \qquad H_0 :\ \delta = 0$$

Also: the paper's strategy return $\eta_t = \operatorname{sign}(ROD_t) \cdot LH_t$ has the same mean on agree and disagree days.

Finding 1: no intraday momentum in post-2020 CL

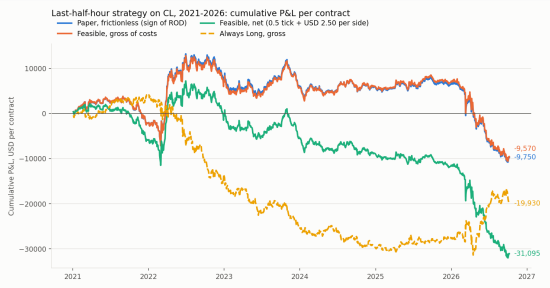


- ROD slope: -1.46, 95% interval $[-3.35, 0.43]$. The paper's CL estimate was $+0.67$ ($t = 0.25$).
- Every post-2020 interval includes zero. The paper's strong results come from other markets and pooled data.

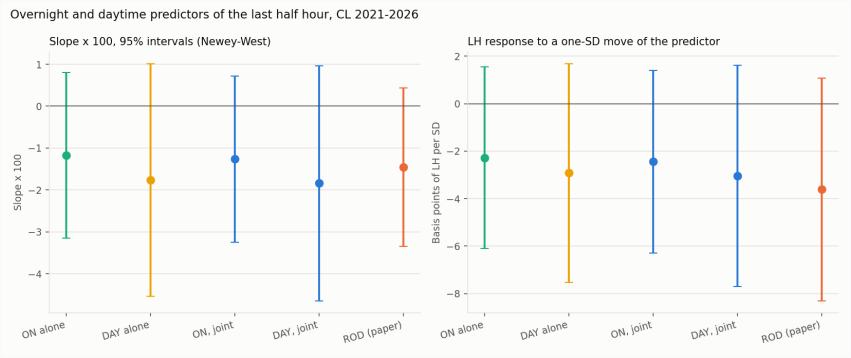
Finding 2: the signal cannot be traded

Long if ROD > 0, short otherwise, over the last half hour.

- Gross: -1.2 bp/day (SE 1.3)
- Net of USD 15 per round trip: -3.1 bp/day, Sharpe -1.1
- Gross is already negative, so no cost level helps

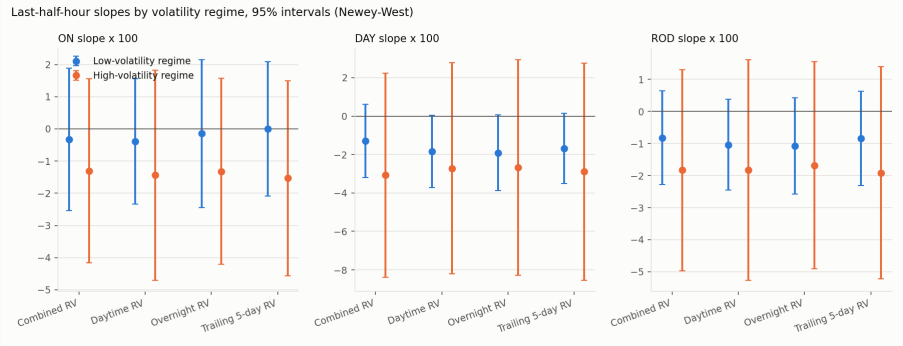


Finding 3: overnight and daytime do not predict differently



- ON slope −1.26 (1.01), DAY slope −1.84 (1.43). Difference 0.58 (1.43).
- Every interval includes zero, and the intervals overlap heavily.

Finding 4: high-volatility regimes change nothing measurable



- High minus low volatility: ON −1.0 (1.8), DAY −1.8 (2.9). The errors are larger than the shifts.
- Realized variance from one-minute returns. High volatility = above the 75th percentile of earlier days.
- Four variance definitions give the same picture.

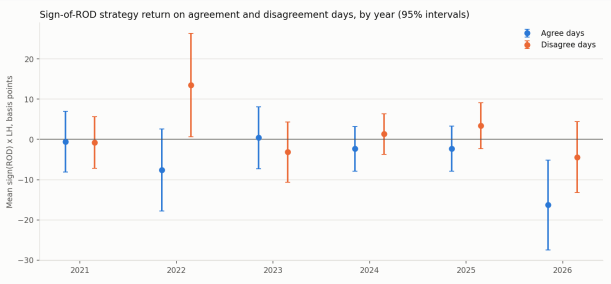
Finding 5: a lead, reversal when ON and DAY agree

Agree: ON and DAY have the same sign (720 days). **Disagree**: opposite signs (704 days).

Paper's strategy return, $\mathrm{sign}(ROD) \cdot LH$:

- Agree: -4.5 bp (1.7), the close reverses
- Disagree: $+1.9$ bp (1.7)

Comes almost entirely from 2022 and 2026. Out-of-sample support rests on one day (10 March 2026, see appendix).



Takeaways and limits

- The paper's momentum result does not show up in CL after 2020, and it was weak for CL in the original sample too.
- The negative slopes are not distinguishable from zero. We do not know why they lean negative.
- One market, six years, two crisis episodes, no quote data (spread is a proxy).
- **Next:** a magnitude-based agreement score, 2019–2020 minute data, other energy futures.

Reproduce: see the README. No API key needed.

Appendix: one day drives much of the signal

10 March 2026, Strait of Hormuz closure: ROD -15.4%, then LH $+4.6\%$.

- Dropping it moves the ROD slope from -1.46 to -0.67 (0.66).
- ROD's only positive out-of-sample R^2 (0.37%) becomes -0.22%.
- The agreement model's out-of-sample R^2 falls from 0.82% to -0.47%.

The data is genuine and stays in every baseline. We report results with and without it.

Appendix: the three-predictor regression

$$LH_t = \alpha + \beta_1\, ONFH_t + \beta_2\, M_t + \beta_3\, SLH_t + \varepsilon_t$$

Slopes $\times 100$ with 95% intervals. The paper's intervals are implied from its slopes and t-statistics.

	Paper, CL 1987–2020	Ours, CL 2021–2026
ONFH	2.40 [−1.80, 6.60]	−1.32 [−3.15, 0.51]
M	−0.06 [−3.98, 3.86]	−1.94 [−5.58, 1.70]
SLH	−13.95 [−41.57, 13.67]	0.38 [−9.33, 10.09]
Adjusted R^2	1.25%	0.46%
Out-of-sample R^2	−1.77%	−1.49%

Every post-2020 interval includes zero. SLH is very noisy in both samples.

Appendix H1: overnight vs. daytime

Slopes $\times 100$, Newey-West standard errors in parentheses. $n = 1{,}436$.

Model	Term	Slope (SE)	95% interval	OOS R^2
$LH \sim ON$	ON	-1.18 (1.01)	$[-3.15,\ 0.80]$	-0.50%
$LH \sim DAY$	DAY	-1.76 (1.42)	$[-4.53,\ 1.02]$	0.12%
$LH \sim ON + DAY$	ON	-1.26 (1.01)	$[-3.25,\ 0.72]$	-0.29%
	DAY	-1.84 (1.43)	$[-4.64,\ 0.97]$	
$LH \sim ROD$ (paper)	ROD	-1.46 (0.97)	$[-3.35,\ 0.43]$	0.37%

- Equality of slopes: $\beta_{ON} - \beta_{DAY} = 0.58$ (1.43), $p = 0.69$.
- Splitting ROD adds no in-sample fit: adjusted R^2 is 0.52% for both the joint model and ROD.
- The daytime part carries the little information there is, mostly in 2026 (DAY slope -9.7 (4.0) in 2026).
- Dropping 10 March 2026 shrinks both slopes to about -0.7.

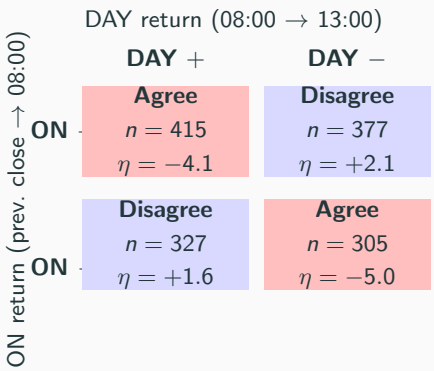
Appendix H2: volatility regimes

High volatility = realized variance above the upper quartile of earlier days. 336 of 1,186 dates from 2022 qualify, mostly in 2022 and 2026. LH standard deviation is 0.70% on those days against 0.37%.

Predictor	Low-vol slope	High-vol slope	High − low (SE), p	Wald p
ON	−0.33 (1.13)	−1.30 (1.46)	−0.97 (1.84), 0.60	0.78
DAY	−1.29 (0.97)	−3.07 (2.71)	−1.79 (2.94), 0.54	
ROD	−0.82 (0.75)	−1.83 (1.60)	−1.01 (1.77), 0.57	

- Daytime, overnight and trailing 5-day variance regimes give the same picture. None of the eight interaction p-values is below 0.41.
- Out of sample, adding regime terms to $LH \sim ON + DAY$ lowers OOS R^2 from 0.35% to between −0.6% and −1.2%, mostly in March 2026.
- Without 10 March 2026 the high-regime differences shrink to −0.07 and +0.70.

Appendix H3: what “agree” and “disagree” mean



Agree: ON and DAY have the same sign, so the day moved one way.

Disagree: opposite signs, so the overnight move was partly undone.

$\eta = \mathrm{sign}(ROD) \cdot LH$, in basis points:

- $\eta > 0$: the close *continues* the day's direction (momentum)
- $\eta < 0$: the close *reverses* it

Example: ON +1% and DAY +0.8% is an agreement day. Momentum predicts LH goes up. On agreement days the close tends to fall instead. Both signs are known at 13:00.

Appendix H3: agreement between ON and DAY

Strategy return $\eta = \mathrm{sign}(ROD) \cdot LH$ in basis points, Newey-West standard errors. 12 days with a zero return are set aside.

Group	n	Mean LH (SE)	η (SE)	Hit rate
ON+/DAY+	415	−4.11 (1.94)	−4.11 (1.94)	0.477
ON−/DAY−	305	5.00 (3.01)	−5.00 (3.01)	0.470
ON+/DAY−	377	−3.69 (2.30)	2.10 (2.34)	0.515
ON−/DAY+	327	−1.07 (2.42)	1.62 (2.39)	0.476
Agree − disagree	1,424		−6.36 (2.34), $p = 0.007$	

- $LH = \alpha + \beta ROD + \gamma Agree + \delta ROD{\cdot}Agree$: $\hat{\beta} = 2.29$ (1.25), $\hat{\delta} = -4.93$ (1.61). The slope is about −2.64 on agreement days.
- Year by year: −21 (8) bp in 2022 and −12 (7) in 2026, within ±6 of zero elsewhere. Without both years: −1.4 (2.3).
- OOS R^2 rises from 0.43% to 0.82% ($p = 0.051$) but is −0.47% without 10 March 2026.